PORTALDOT HACKER HOUSE 2026 · ECOSYSTEM DAPPS

PortaStake Shield

Liquid staking savings with built-in slashing insurance, live on Portaldot V3 testnet.

Team: Aloys Jehwin and Clatson

THE PROBLEM

Staking yield is hard to use safely

Pick a validator

A choice most savers cannot judge.

Carry the slash

A validator penalty lands on the staker.

Wait ~168 days

Unbonding takes 672 eras on Portaldot.

So savers, DAO treasuries and NGO reserves leave their capital idle.

THE SOLUTION

One deposit: liquid, earning, insured

Deposit

tPOT, tUSD or xUSD into one ERC-4626 vault.

Hold stPOT

A liquid share, plus a soulbound Policy NFT.

Earn

RateEngine targets 8% and funds insurance from surplus.

Protect

Attested slashes are covered up to the fund’s balance.

Around it: autopilot savings mandates, instant exits from a hot reserve, a USDC bridge, x402 paid APIs and an SDK.

WHY PORTALDOT IS ESSENTIAL

Native staking and an EVM on one network

Substrate

Native staking, and the yield source.

Relayer

Connects the two runtimes for sweep, bond and rewards.

pallet-revive EVM

Solidity vault, insurance and automation.

No staking precompile, so the product is designed around Portaldot’s architecture.

~168-day unbonding is what makes a liquid share and a hot reserve valuable here.

PROOF OF FUNCTION

Check every step on the explorer

Deposit	5 tPOT into the vault, stPOT minted	0xf2e2…13bf
Withdraw	5 stPOT redeemed for 2.49 tUSD, no waiting	0x68a2…da62
Mandate control	Owner paused a savings rule on-chain	0x9f4f…6b5d
x402 payment	0.01 xUSD settled, gas paid by the facilitator	0x74ed…13b6
Bridge, source side	5 USDC sent from Base Sepolia	0xcb7d…128c

PROOF OF FUNCTION

A working product, not a mockup

11
contracts deployed on V3 testnet

90
unit, fuzz and invariant tests

7
pages in the live app

1
SDK for x402 and the protocol

The API gateway turned NASA’s public API into a paid x402 endpoint, with its key sealed, and the test call returned **200**.

WHERE IT STANDS

What is live and what comes next

Live on testnet

- Vault, deposits, instant exits
- Mandates and keeper
- x402 tools and API gateway

Built, still proving

- Bridge: tested on local chains
- Relayer: single key today

Not done yet

- Security audit
- Mainnet launch

WHY CONTINUED SUPPORT

What Portaldot gains as this grows

A yield-bearing base asset

stPOT can become collateral for future DeFi apps.

Stake for new validators

Insurance gives cautious capital a reason to delegate.

Recurring on-chain activity

Deposits, mandate runs and per-call x402 payments.

Tools for other builders

An SDK, a facilitator and a gateway any app can use.

THE PATH TO MAINNET

What support would unlock

Step 1

Live staking

Fund the stash and run the daemon on testnet.

Step 2

Less trust

Move from one relayer key to a 3-of-5 multisig.

Step 3

Audit

Review the contracts and relayer, then publish the SDK.

Step 4

Mainnet

Launch with validator partners and first treasuries.

Our ask: testnet POT for the stash, technical access to the Portaldot team, and help with audit and validator introductions.

Safer yield and real on-chain activity for Portaldot

portastake-shield.vercel.app

github.com/AloysJehwin/portastake-shield

Demo video: [public video link]